

URL Shortener — Configuration and Environment

Project: URL Shortener

Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic

Location: /home/dominic/Documents/fastapiurl

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Environment Variables

```
DATABASE_URL=sqlite:///./urls.db
```

Minimal configuration. No additional config files.

Configuration & Environment

Environment Files

```
# .env (example)
DATABASE_URL=sqlite:///./data.db
SECRET_KEY=your-secret-key-here
DEBUG=false
CORS_ORIGINS=*
```

Configuration Loading

```
# Pattern used across projects
from pydantic_settings import BaseSettings

class Settings(BaseSettings):
    database_url: str = "sqlite:///./data.db"
    secret_key: str = "change-me"
    debug: bool = False
    cors_origins: str = "*"

    model_config = {"env_file": ".env", "env_file_encoding": "utf-8"}
```

Runtime Configuration

Parameter	Source	Default	Description
DATABASE_URL	.env	sqlite:///./data.db	DB connection
SECRET_KEY	.env	(required)	JWT signing
DEBUG	.env	false	Dev mode
CORS_ORIGINS	.env	*	CORS policy

Deployment-Specific Config

Render Web Service (free tier): - CPU: Shared - RAM: 512 MB - Disk: 1 GB - Region: Frankfurt - Auto-deploy: On push to main

Sensitive Values

Variable	Risk	Mitigation
SECRET_KEY	JWT forgery	Use strong random value, rotate periodically
OPENAI_API_KEY	Cost, data exposure	Server-side only, rate-limited
DATABASE_URL	Data access	Never commit to git